LibNet Case Study

LibNet is a public library system operating across multiple branches in a metropolitan area. The library network lends physical media (books, audiobooks, DVDs), manages patron memberships, hosts community events, and tracks staff operations across branches.

Currently, each branch operates using legacy desktop files, creating problems with lost inventory, untracked overdue fees, and conflicting event bookings. Your team has been hired as database consultants to design a unified relational database schema that handles multi-branch library operations seamlessly.

Th system focuses on the following domain logic:

- **Branches & Facilities:** The network consists of multiple library branches. Each branch has a name, address, phone number, and operating hours.
- **Books & Titles:** The library maintains a central catalog of book titles. Each title record includes an ISBN, title, publisher, publication year, and primary genre.
- **Physical Copies:** A single book title can have multiple physical copies distributed across different branches. Each copy has a unique barcode, acquisition date, shelf location tag, and current condition status (e.g., *Available*, *Checked Out*, *Damaged*, *Under Maintenance*).
- **Authors:** Books are written by authors. The system tracks author details (first name, last name, biography, nationality). A title can be written by multiple authors, and an author can write multiple titles.
- **Patrons:** Local residents register as library patrons. Each patron record tracks their library card number, full name, email, contact phone, residential address, registration date, and current account status (e.g., *Active*, *Suspended*).
- **Loans & Borrowing:** Patrons check out specific physical copies from a branch. Each loan record logs the issue date, due date, actual return date, and the staff member who processed the loan.
- **Fines & Payments:** Overdue or damaged returns incur fines. A fine is tied to a specific loan transaction and records the fine amount, reason, date issued, payment date, and payment status (e.g., *Pending*, *Paid*, *Waived*).
- **Community Events:** Branches host public events (e.g., *Book Clubs*, *Children's Storytime*, *Digital Literacy Workshops*). Event records include the event name, description, scheduled date/time, maximum capacity, target age group, and assigned host branch.